Linux Task Documentation – User, Group & File Permissions

Date: 9 February 2026

Objective:

The objective of this task was to understand and practice Linux user management, group management, file creation, and permission handling using terminal commands.

Environment:

- OS: Linux (Ubuntu / WSL)

- Terminal: Bash

- Users: shariqua, elon

Tasks Performed:

- Created and verified Linux groups

- Added user to groups and sudo

- Switched users

- Created files

- Checked and modified file permissions

Steps Performed:

1. Created Groups:

sudo addgroup devteam

sudo addgroup devteams

2. Added User to Groups:

sudo usermod -aG devteams elon

sudo usermod -aG sudo elon

3. Verified Groups:

groups elon

4. Switched User:

su elon

5. Created File:

touch project.txt

6. Checked Permissions:

ls -l

7. Changed Permissions:

chmod 640 project.txt

Permission Explanation:

6 = Read + Write (Owner)

4 = Read only (Group)

0 = No access (Others)

Final Permission:

-rw-r----- 1 elon elon project.txt

```
elon@DESKTOP-8UVUP6Q: ~
elon:x:1001:1001:elon,,,:/home/elon:/bin/bash
shariqua@DESKTOP-8UVUP6Q:~$ sudo addgroup devteam
addgroup: The group `devteam' already exists.
shariqua@DESKTOP-8UVUP6Q:~$ sudo addgroup devteams
Adding group `devteams' (GID 1002) ...
Done.
shariqua@DESKTOP-8UVUP6Q:~$ sudo usermod -aG devteams elon
shariqua@DESKTOP-8UVUP6Q:~$ groups elon
elon : elon devteams
shariqua@DESKTOP-8UVUP6Q:~$ sudo usermod -aG sudo elon
shariqua@DESKTOP-8UVUP6Q:~$ groups elon
elon : elon sudo devteams
shariqua@DESKTOP-8UVUP6Q:~$ su elon
Password:
su: Authentication failure
shariqua@DESKTOP-8UVUP6Q:~$ su elon
Password:
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

elon@DESKTOP-8UVUP6Q:/home/shariqua$ cd
elon@DESKTOP-8UVUP6Q:~$ touch poject.txt
elon@DESKTOP-8UVUP6Q:~$ ls -l
total 0
-rw-rw-r-- 1 elon elon 0 Feb  9 01:09 poject.txt
elon@DESKTOP-8UVUP6Q:~$ chmod 640 poject.txt
elon@DESKTOP-8UVUP6Q:~$ ls -l
total 0
-rw-r----- 1 elon elon 0 Feb  9 01:09 poject.txt
elon@DESKTOP-8UVUP6Q:~$
```

Conclusion:

This task provided hands-on experience with Linux user, group, and permission management.

Author:

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